

Automatic Grid-Following $\leftrightarrow$ Grid-Forming Mode Switching of Two IBRs via an Adaptive Grid Stability Index (AGSI) on an Infinite Bus

Principle, switching equation, and current-continuous (bumpless) transfer

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Abstract

This report (separate from the earlier 6-state GFL/GFM comparison report) presents the **automatic mode-switching supervisor** that toggles two IBRs between grid-following (GFL) and grid-forming (GFM) on a common point of connection (PCC) behind one line to an ideal infinite bus. The decision uses a *single index equation*, the **Adaptive Grid Stability Index (AGSI)** (EECON49-P4 guideline), compared with two *reference lines* ($\Gamma_{on}, \Gamma_{off}$) plus a hysteresis band and dwell times. A temporary weak-grid event drives the AGSI of both units up to Γ_{on} (switch GFL→GFM); on recovery the AGSI falls below Γ_{off} (switch back GFM→GFL). Every transfer is *current-continuous* (bumpless). Project-owned time-domain simulation (no external solver) confirms a converged bidirectional cycle, current continuity at the 10^{-17} level, and passing automated tests. All equations and thresholds are traceable; nothing was tuned to bias the result.

1 Scope and link to the previous report

The previous report (`report_ibr_gfl_gfm_en`) documented the two reduced 6-state IBR models (GFL, GFM) and their SSSA. Here those models are *reused as the single source of truth* for the branch equations, and a **supervisor** that switches modes automatically is added. Source discipline: the switching structure (AGSI, thresholds, dwell) follows the EECON49-P4 strategy [1] as a *design guideline, not an exact reproduction* (the paper’s Bayesian Optimization is not used here); classification `PROJECT_DERIVED_SOURCE_GUIDED`. Bidirectional switching and state transfer are consistent with the switchable-control literature [2, 3, 4].

2 Principle and test system

GFL relies on a PLL and is well behaved in strong grids but loses damping / stability in weak grids; GFM forms its own voltage/frequency reference and rides through weak grids. The supervisor therefore switches GFL→GFM when the system is stressed and GFM→GFL when it recovers. **The disturbance that forces the switch** here is a temporary *grid-weakening event*: at t_{ev} the line impedance to the infinite bus is multiplied ($\times 4$ default, $Z_{line} : 0.30j \rightarrow 1.20j$), i.e. the SCR drops to $\sim 1/4$ (modelling loss of a strong transmission path); the framework also supports a V_∞ phase/voltage step and loss of reference (GRA = 0, islanding). Causal chain: a GFL is a current source, so through the high-impedance line the PCC voltage sags ($|V| : 0.99 \rightarrow 0.79$), which sharply raises J_V ; near the max-power (voltage-collapse) nose the PLL loses damping, adding J_f, J_R ; the weighted AGSI then crosses Γ_{on} and the units switch to GFM (which forms voltage + virtual inertia and rides through the weak grid). On line restoration the grid is strong again, AGSI falls below Γ_{off} , and the units return to GFL. Two IBRs start as GFL at a common PCC; the PCC algebraic balance is

$$I_1(x_1, V) + I_2(x_2, V) = \frac{V - V_\infty}{Z_{line}}. \quad (1)$$

Each device is an object (`ibr.SwitchableIbr6`) wrapping both 6-state branches; the active state dimension is always 6, and a switch re-initialises those 6 slots to the target-branch equilibrium.

3 The switching equation (AGSI) and how the index acts

The AGSI merges four normalised PCC deviations into one dimensionless index:

$$\text{AGSI} = w_V J_V + w_f J_f + w_R J_R + w_P J_P, \quad w_V + w_f + w_R + w_P = 1, \quad (2)$$

$$J_V = \frac{|V - V_{ref}|}{\Delta V_{base}}, \quad J_f = \frac{|f - f_0|}{\Delta f_{base}}, \quad J_R = \frac{|df/dt|}{\Delta \text{RoCoF}_{base}}, \quad J_P = \frac{|P_{ref} - P|}{\Delta P_{base}}. \quad (3)$$

In GFL f is the PLL frequency; in GFM it is the VSG rotor frequency; df/dt is a one-per-step numerical RoCoF. Each term is a different “stress view” of the PCC:

- J_V — **voltage deviation**: V = measured PCC voltage vs nominal $V_{ref} = 1$ pu; a weak grid / heavy loading sags V (the dominant driver here). $\Delta V_{base} = 0.10$ pu ($10\% \Rightarrow J_V = 1$).
- J_f — **frequency deviation**: how far f strays from $f_0 = 60$ Hz; indicates power imbalance / loss-of-synchronism tendency. $\Delta f_{base} = 0.50$ Hz.
- J_R — **RoCoF**: rate of change of frequency, a standard low-inertia stability metric [5]; large RoCoF $\Rightarrow$ low effective inertia $\Rightarrow$ favour GFM (virtual inertia). $\Delta \text{RoCoF}_{base} = 1.0$ Hz/s (grid-code scale ~ 0.5 – 2 Hz/s).

- J_P — **active-power tracking error**: commanded P_{ref} vs delivered P ; a large error means the converter cannot meet its command (weak grid / current limit). $\Delta P_{base} = 0.20$ pu.
- weights $w_\bullet$ (sum 1) set the relative importance; the normalisation bases make the terms dimensionless and comparable (a deviation equal to its base gives sub-index 1); Γ_{on}/Γ_{off} are the AGSI thresholds; GRA flags whether a voltage/frequency reference exists; T_d is the dwell that rejects momentary spikes.

With grid-reference availability $GRA \in \{0, 1\}$ the logic is:

$$\text{GFL} \rightarrow \text{GFM} : (\text{AGSI} \geq \Gamma_{on}) \vee (\text{GRA} = 0) \text{ held} \geq T_{d,on}, \quad (4)$$

$$\text{GFM} \rightarrow \text{GFL} : (\text{AGSI} < \Gamma_{off}) \wedge (\text{GRA} = 1) \text{ held} \geq T_{d,off}. \quad (5)$$

Since $\Gamma_{on} > \Gamma_{off}$, the band $[\Gamma_{off}, \Gamma_{on}]$ is a *hysteresis* region with no switching; the dwell times reject momentary transients (e.g. a brief RoCoF spike at a topology change). Defaults (EECON49-P4 §4.1 guideline): $w = [0.30, 0.30, 0.25, 0.15]$; $\Delta V_{base} = 0.10$ pu, $\Delta f_{base} = 0.50$ Hz, $\Delta \text{RoCoF}_{base} = 1.0$ Hz/s, $\Delta P_{base} = 0.20$ pu; $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$; $T_{d,on} = 0.10$ s, $T_{d,off} = 1.00$ s.

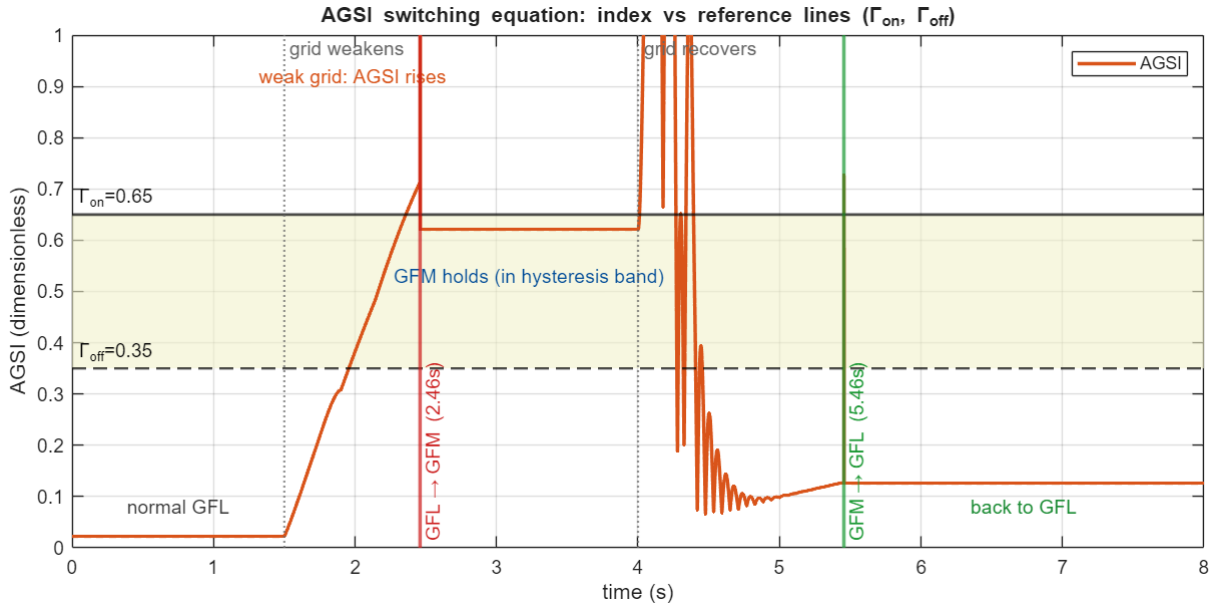


Figure 1: AGSI versus the two reference lines (yellow = hysteresis band). The index rises to Γ_{on} (switch to GFM, red), holds in the band while the grid is weak, then falls below Γ_{off} after recovery (switch back to GFL, green).

4 Bumpless transfer

At the switch instant, the terminal (V, P, Q) is read from the active injection and the target branch is initialised to the equilibrium delivering the same (P, Q) at the same V . Because both branches give $I = \overline{(P + jQ)/V}$, the injected current is continuous ($|I^+ - I^-| \approx 2.8 \times 10^{-17}$ in both directions). The GFM references are back-solved so its swing/voltage derivatives are also zero at the instant.

5 Results

Figure 2 shows the default scenario (line weakened $\times 4$ over 1.5–4.0 s, smooth 0.4 s ramp, no phase snap): normal GFL $\rightarrow$ AGSI rises $\rightarrow$ GFL $\rightarrow$ GFM at ≈ 2.46 s $\rightarrow$ GFM holds the weak grid (AGSI in the hysteresis band) $\rightarrow$ grid recovers $\rightarrow$ GFM $\rightarrow$ GFL at ≈ 5.45 s $\rightarrow$ normal GFL. Verified: two transitions per unit ending in GFL; Newton converges at every step; current continuous to $\sim 10^{-17}$; frequency swing only ± 0.6 Hz; pre-event equilibrium residual at machine precision; all tests in `tests/test_ibr_switchable6.m` pass (including the `solve_case` route).

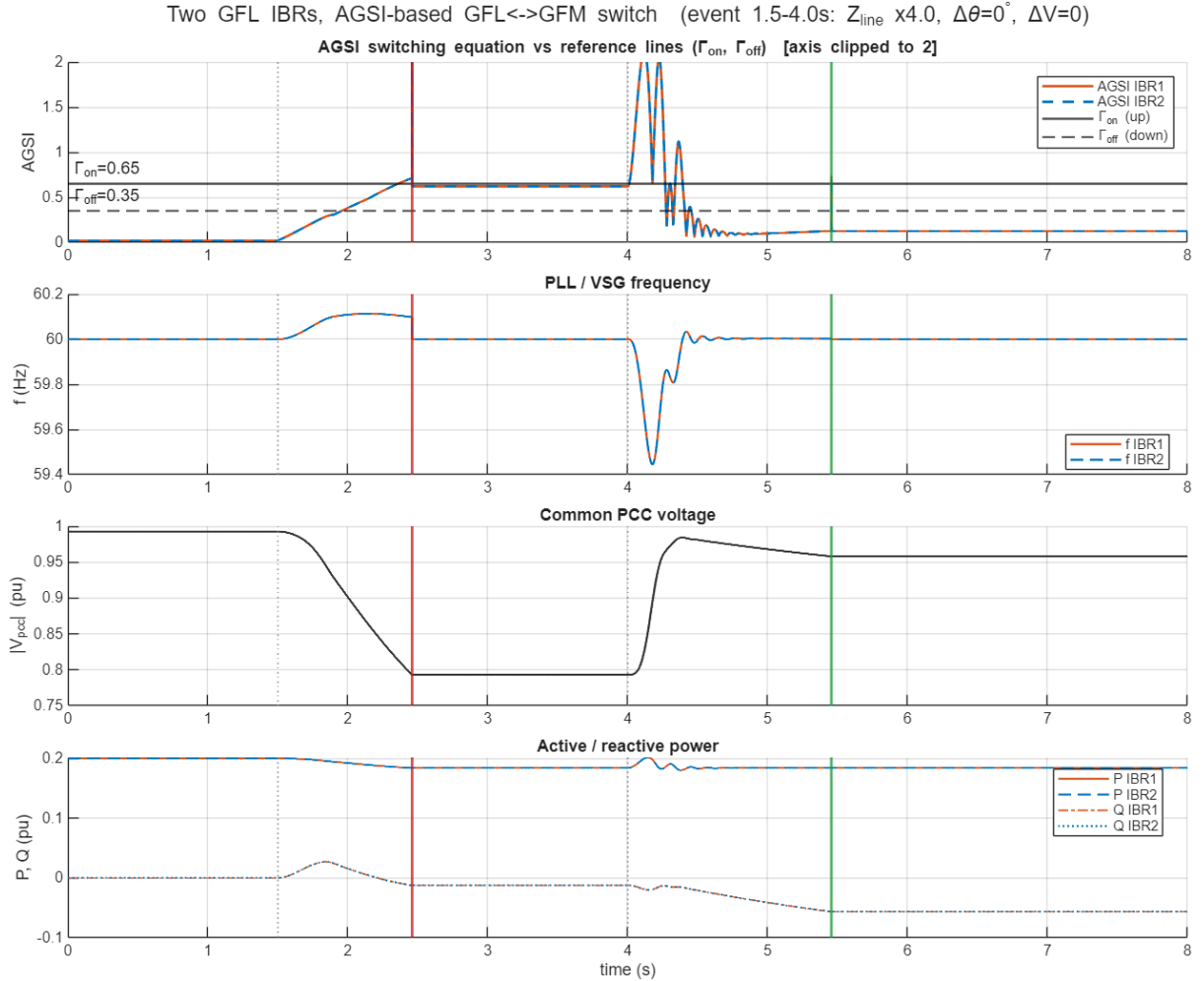


Figure 2: Four-panel overview: AGSI vs reference lines (red = GFL $\rightarrow$ GFM, green = GFM $\rightarrow$ GFL), PLL/VSG frequency, PCC voltage, and P, Q of both IBRs.

6 How to run

`run_two_ibr_switch` (opens a settings dialog, then runs) • `run_two_ibr_switch(Zline_factor=5)` (skip the dialog) • `solve_case('analysis', 'ibr', 'case', 'two_ibr_switch')`.

7 Limitations

Weights/thresholds are PROJECT_DERIVED_SOURCE_GUIDED (no Bayesian Optimization); at a common PCC both IBRs see the same voltage and switch together (a separate-bus network is the next step); the driver is a diagnostic, not a production TS kernel; no current limiter/LVRT. The result is structural evidence of the switching mechanism, not a fault-ride-through certification.

8 Extended study: automatic switching on two multi-machine systems

This section extends the single-PCC demonstration to two multi-machine systems in which one synchronous generator (SG) shares the network with several grid-following (GFL) IBRs. On an SG trip the stressed IBRs form (GFL→GFM) to hold the island; on a synchronised SG reclose the SG re-takes the slack and the IBRs hand the reference back. The decision uses the same AGSI supervisor, upgraded here to **AGSI++**.

8.1 AGSI / AGSI++ switching index (corrected)

The Adaptive Grid-Support Index is a weighted sum of normalised sub-indices, evaluated per IBR every step:

$$J_V = \frac{|V - V_{ref}|}{\Delta V_{base}}, \quad J_f = \frac{|f - f_0|}{\Delta f_{base}}, \quad (6)$$

$$J_R = \frac{|df/dt|}{\Delta R_{base}}, \quad J_P = \frac{|P_{ref} - P|}{\Delta P_{base}}, \quad (7)$$

$$J_{SCR} = \max\left(0, \frac{SCR_{crit}}{SCR} - 1\right), \quad J_{lock} = \frac{|v_q|}{\Delta v_{q,base}}. \quad (8)$$

with $\Delta V_{base} = 0.10$ pu, $V_{ref} = 1.0$; $\Delta f_{base} = 0.50$ Hz, $f_0 = 60$; $\Delta R_{base} = 1.0$ Hz/s (RoCoF); $\Delta P_{base} = 0.20$ pu; $SCR_{crit} = 3$; and $\Delta v_{q,base} = 0.10$. The composite index is

$$AGSI = w_V J_V + w_f J_f + w_R J_R + w_P J_P + w_{SCR} J_{SCR} + w_{lock} J_{lock}. \quad (9)$$

Plain AGSI uses $[w_V, w_f, w_R, w_P] = [0.30, 0.30, 0.25, 0.15]$ (with $w_{SCR} = w_{lock} = 0$); AGSI++ uses $[0.25, 0.25, 0.15, 0.10, 0.15, 0.10]$ (sum = 1). Switching applies hysteresis and dwell:

$$\text{GFL} \rightarrow \text{GFM} : \quad AGSI \geq \Gamma_{on} = 0.65 \text{ held for } \geq T_{d,on}, \quad (10)$$

$$\text{GFM} \rightarrow \text{GFL} : \quad AGSI < \Gamma_{off} = 0.35 \text{ held for } \geq T_{d,off}. \quad (11)$$

AGSI++ low-pass filters the RoCoF term ($\tau_R = 0.05$ s) to reject the one-sample RoCoF spike that causes no-dwell chattering, and adds J_{SCR} (weak-grid strength) and J_{lock} (a PLL loss-of-lock proxy using $|v_q|$).

8.2 Reduced-6 IBR models (state order + equations)

Each functional block contributes two dynamic states; the input is $u = [P_{ref}; Q_{ref}]$ on system base. **GFL (grid-following)** has state order $x = [i_d, i_q, \delta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q]$ (dq L-filter currents on inverter base, SRF-PLL angle and PI integrator, and the P/Q outer-PI integrators):

$$\dot{\delta}_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}, \quad \dot{\xi}_{PLL} = v_q, \quad (12)$$

$$\dot{\xi}_P = e_P = P_{ref} - P, \quad \dot{\xi}_Q = e_Q = Q_{ref} - Q, \quad (13)$$

$$i_{d,ref} = k_{p,P} e_P + k_{i,P} \xi_P, \quad i_{q,ref} = -(k_{p,Q} e_Q + k_{i,Q} \xi_Q), \quad (14)$$

$$\dot{i}_d = \frac{\omega L i_q - R_t i_d + v_{td} - v_d}{L/\omega_b}, \quad \dot{i}_q = \frac{-\omega L i_d - R_t i_q + v_{tq} - v_q}{L/\omega_b}. \quad (15)$$

GFM (grid-forming, no PLL) has state order $x = [i_d, i_q, \omega, \delta, E, \xi_V]$ (currents, virtual-rotor speed and angle, internal voltage magnitude, and the d-axis voltage-PI integrator):

$$\dot{\omega} = \frac{P_{ref} - P - D_v(\omega - 1)}{M}, \quad \dot{\delta} = \omega_b(\omega - 1), \quad (16)$$

$$\dot{E} = \frac{k_Q(Q_{ref} - Q) - k_E(E - |V|)}{\tau_E}, \quad \dot{\xi}_V = E - v_{gd}, \quad (17)$$

$$i_{q,ref} = -(k_{p,V}(E - v_{gd}) + k_{i,V}\xi_V), \quad i_{d,ref} = k_{p,V}(-v_{gq}), \quad (18)$$

where the terminal quantities are *cross-paired*, $v_{gd} = E + X i_q$ and $v_{gq} = -X i_d$; a same-axis pairing would make the voltage loop unstable.

8.3 SwitchableIbr6 device and bumpless transfer

A single **SwitchableIbr6** device wraps one GFL and one GFM reduced-6 model as the single source of truth; the active state is always 6-dimensional. A mode switch is *bumpless*: the target mode is re-initialised to the instantaneous terminal (V, P, Q) so the injected current is continuous,

$$I^+ = \overline{\left(\frac{S}{V}\right)} = I^-. \quad (19)$$

The AGSI supervisor decides switching per IBR with hysteresis and dwell. On an SG trip the stressed IBRs (AGSI crossing Γ_{on}) form GFL→GFM; on a synchronised SG reclose the SG re-takes the slack (angle reference) and the IBRs hand the reference back and revert to their scheduled GFL dispatch (the index decides which revert).

8.4 Synchronous generator: Padiyar model-1.1

State order $[\delta, \omega, E'_q, E'_d]$ for *manual* (constant field, no AVR, 4-state) or $[\delta, \omega, E'_q, E'_d, E_{fd}]$ for *avr* (5-state):

$$\dot{\delta} = \omega_B(\omega - 1), \quad (20)$$

$$\dot{\omega} = \frac{P_{m,eff} - T_e - D(\omega - 1)}{2H}, \quad P_{m,eff} = P_m - \frac{\omega - 1}{R}, \quad (21)$$

$$\dot{E}'_q = \frac{E_{fd} - E'_q - (X_d - X'_d)I_d}{T'_{d0}}, \quad (22)$$

$$\dot{E}'_d = \frac{-E'_d + (X_q - X'_q)I_q}{T'_{q0}}, \quad (23)$$

$$\dot{E}_{fd} = \frac{K_A(V_{ref} - |V|) - E_{fd}}{T_A} \quad (\text{avr only}). \quad (24)$$

Here $P_{m,eff}$ carries the primary-governor droop ($R = \infty$ disables it); in manual mode $E_{fd} = E_{fd0}$ is constant. The stator solution is

$$r_d = V_d - E'_d, \quad r_q = V_q - E'_q, \quad \text{den} = R_a^2 + X'_d X'_q, \quad (25)$$

$$I_d = \frac{-R_a r_d - X'_q r_q}{\text{den}}, \quad I_q = \frac{X'_d r_d - R_a r_q}{\text{den}}, \quad (26)$$

$$T_e = V_d I_d + V_q I_q + R_a (I_d^2 + I_q^2). \quad (27)$$

8.5 Test system 1: Padiyar two-area (unstable operating point)

One SG at bus 11 (AVR, 5-state) plus three GFL IBRs at buses 1, 2 and 12. The SG-online composite has **23 states** and **one right-half-plane eigenvalue**, $\max \text{Re} = +0.058 \text{ s}^{-1}$, so the operating point is *small-signal unstable*; the least-damped oscillatory mode is $f = 0.677 \text{ Hz}$ with $\zeta = +0.12$. Switching timeline (temporary three-phase fault $Z_f = j0.1 \text{ pu}$ at bus 3 during 1.0–1.15 s, then SG trip at 2 s and reclose at 5 s, over a 50 s window):

1. $[0, 1)$ s: SG is the slack; all three IBRs are GFL.
2. 1.0–1.15 s: temporary fault ($Z_f = j0.1 \text{ pu}$); the IBRs ride through it as GFL.
3. $t = 2$ s: SG trips; all three IBRs cross Γ_{on} and form GFM (island forms).
4. $t = 5$ s: synchronised reclose; the SG re-takes the slack and the IBRs hand the reference back.
5. $t \rightarrow 50$ s: because the operating point is unstable, the slow mode grows and the system collapses to all-GFM with $V_{end} \approx 0.30 \text{ pu}$.

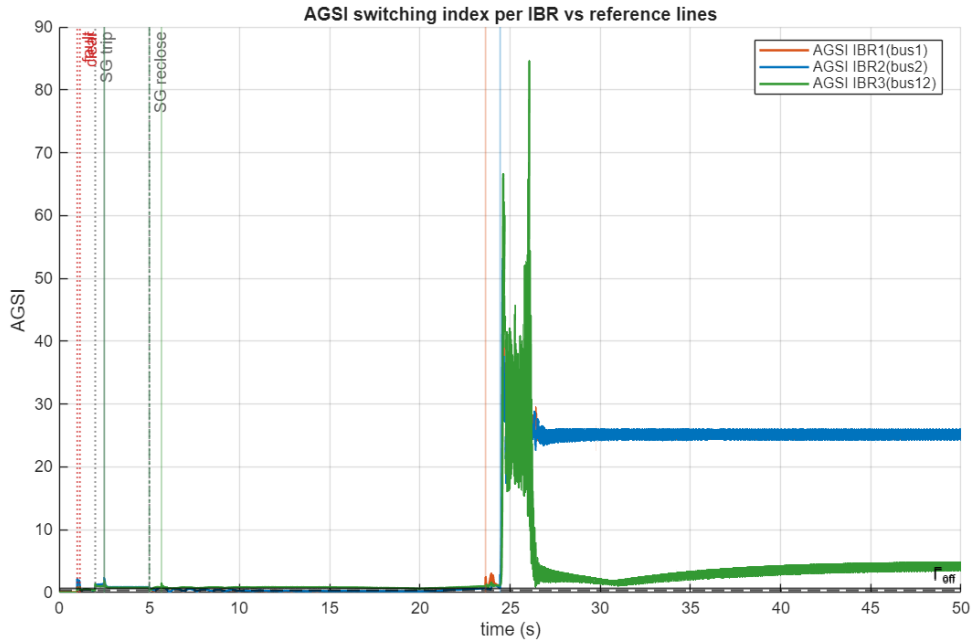


Figure 3: Padiyar two-area: AGSI++ per IBR (fault $Z_f=j0.1 \text{ pu}$, then SG trip 2 s, reclose 5 s; 50 s window).

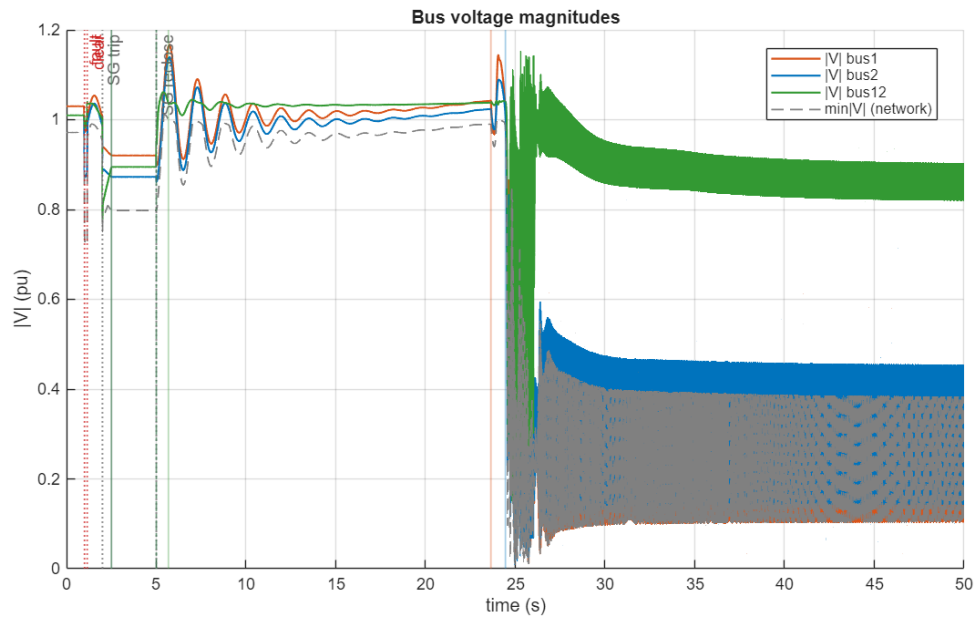


Figure 4: Padiyar two-area: bus voltages through the trip/reclose cycle.

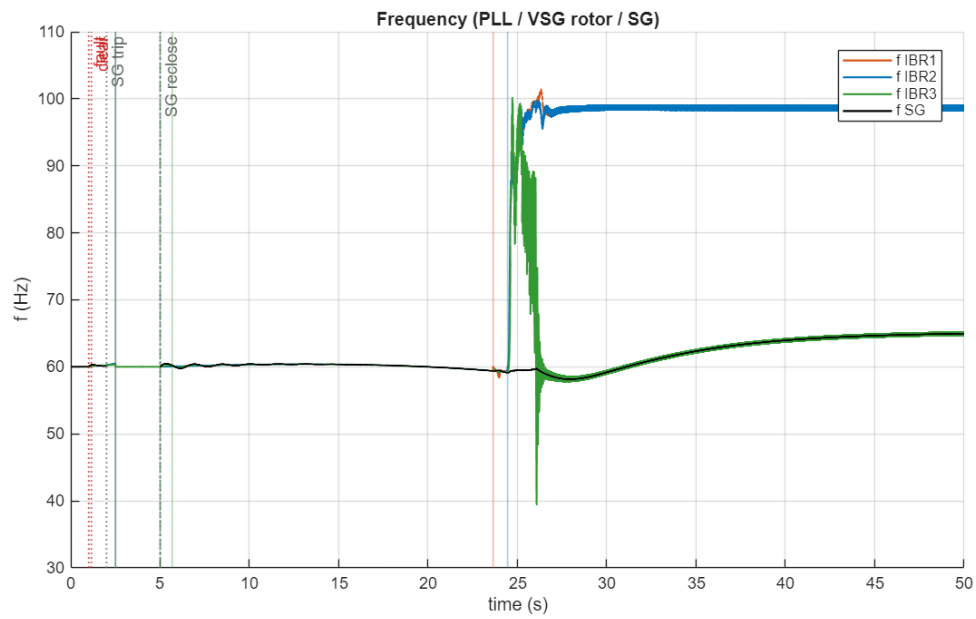


Figure 5: Padiyar two-area: SG and IBR frequencies.

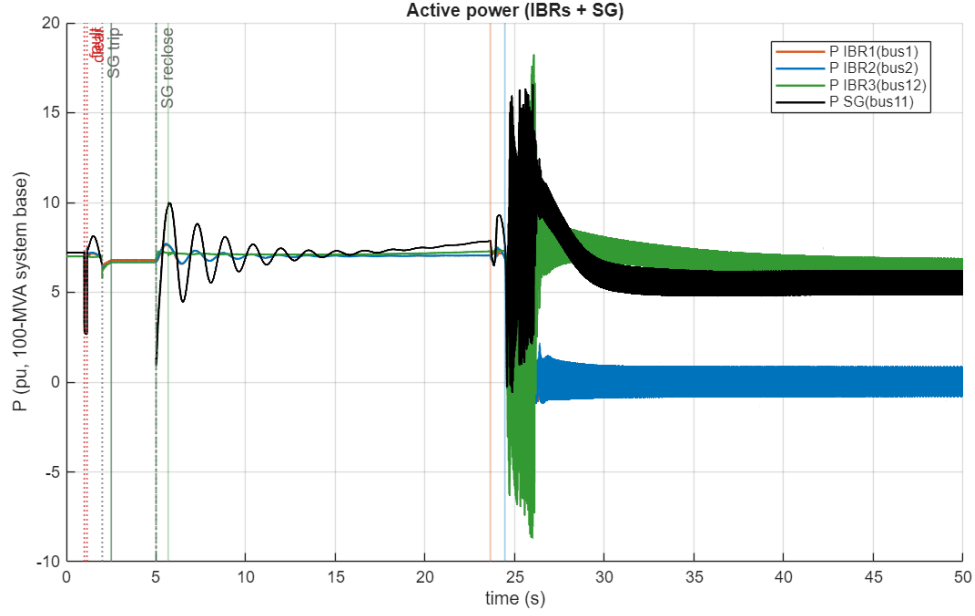


Figure 6: Padiyar two-area: active-power injections.

8.6 Test system 2: IEEE 14-bus (stable operating point)

One SG at bus 1 (manual, no AVR, 4-state) plus four GFL IBRs at buses 2, 3, 6 and 8 (reduced-6). The composite has **28 states** and **zero right-half-plane eigenvalues**, $\max \text{Re} \approx 0$ (only the ≈ 0 angle-reference mode); the least-damped mode is $f = 0.345$ Hz with $\zeta = +0.267$, so the operating point is *small-signal stable*. Switching timeline (temporary three-phase fault $Z_f = j0.1$ pu at bus 4 during 1.0–1.15 s, then SG trip at 2 s and reclose at 5 s, over a 50 s window):

1. $[0, 1)$ s: SG is the slack; all four IBRs are GFL.
2. 1.0–1.15 s: temporary fault ($Z_f = j0.1$ pu); the IBRs ride through it as GFL.
3. $t = 2$ s: SG trips; the stressed IBRs cross Γ_{on} and form GFM (the island survives).
4. $t = 5$ s: synchronised reclose; the angle reference returns to the SG.
5. $t \rightarrow 50$ s: the response stays bounded ($V_{end} \approx 0.94$ pu, most IBRs revert to GFL, AGSI bounded around 3.5).

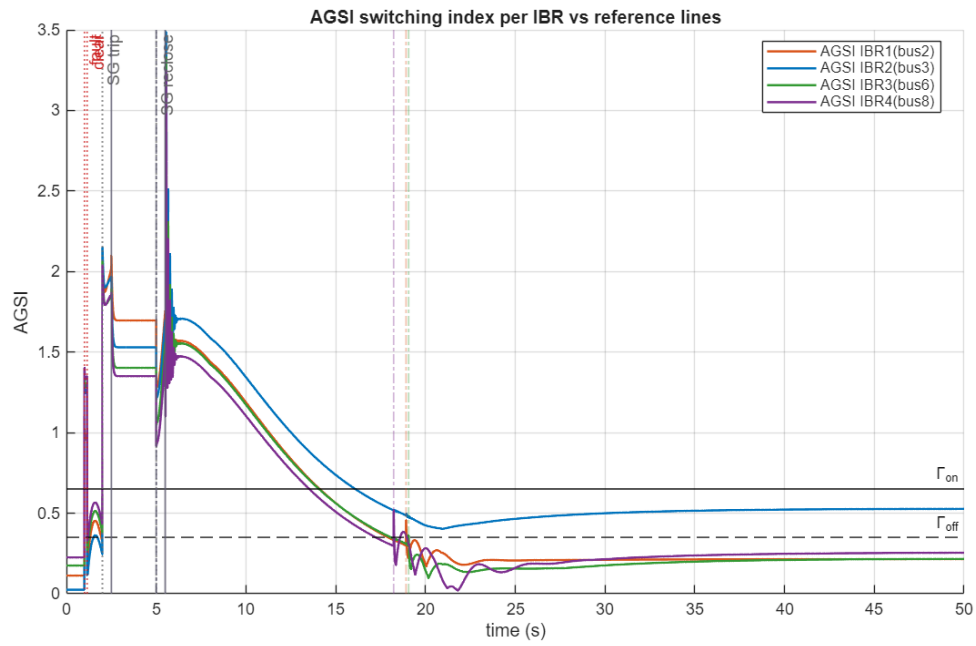


Figure 7: IEEE 14-bus: AGSI++ per IBR (fault $Z_f=j0.1$ pu, then SG trip 2 s, reclose 5 s; 50 s window).

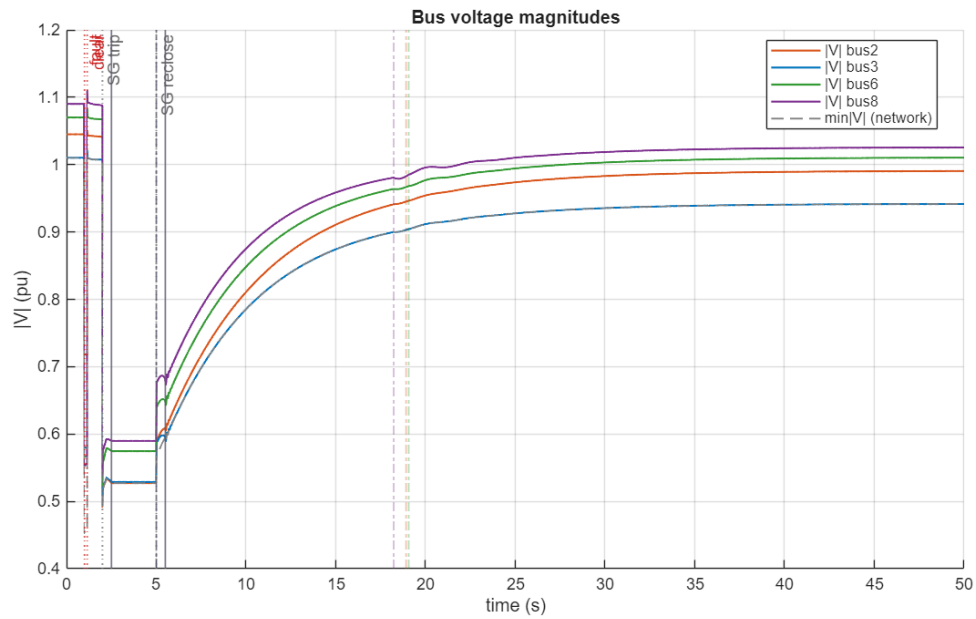


Figure 8: IEEE 14-bus: bus voltages through the trip/reclose cycle.

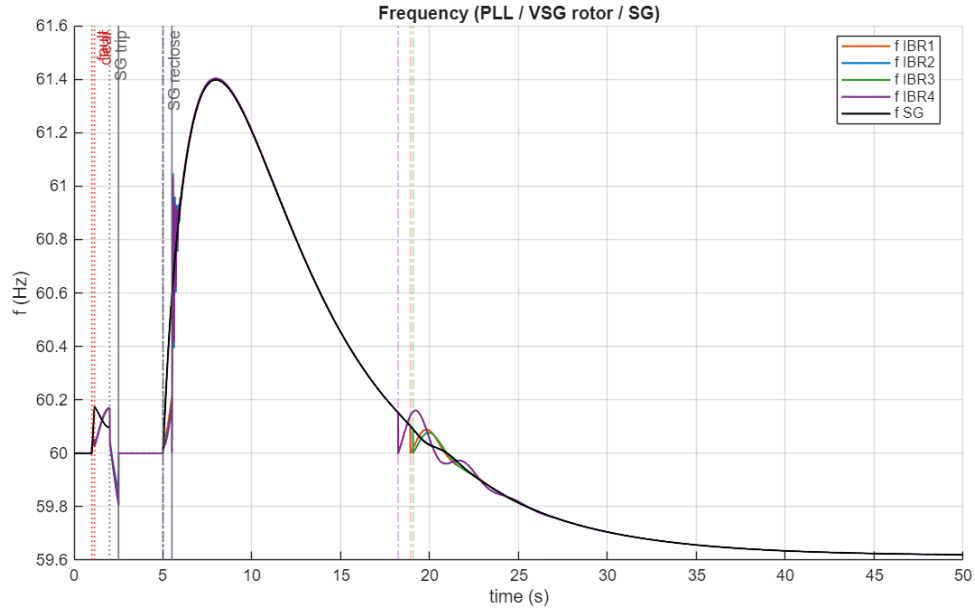


Figure 9: IEEE 14-bus: SG and IBR frequencies.

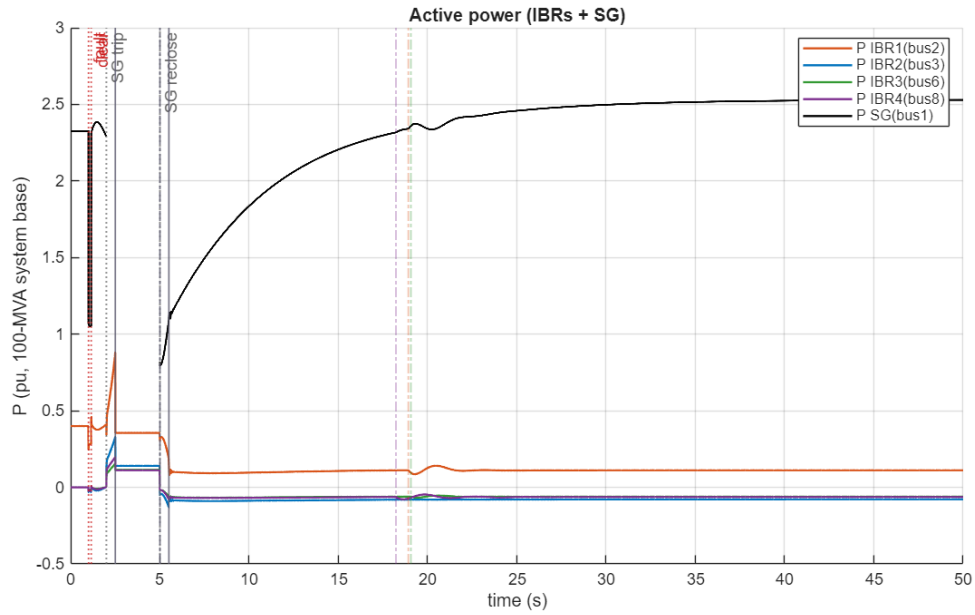


Figure 10: IEEE 14-bus: active-power injections.

8.7 Summary

The two systems share an identical supervisor and event sequence but differ in outcome: the Padiyar two-area operating point is small-signal unstable ($+0.058 \text{ s}^{-1}$) and collapses to all-GFM on a long run, whereas the meshed IEEE 14-bus operating point is stable and stays bounded. AGSI++ keeps the index bounded and non-chattering in both cases.

References

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